An Elementary Algebraic PCP and PCPP

Quadratic arithmetization, multilinearity testing, and tabulated sum-check

August 2026

Abstract

We give a deliberately parameter-wasteful proof that every language in NP has a probabilistically checkable proof using polylogarithmically many random bits and polylogarithmically many queries. The proof has only three ingredients: quadratic arithmetization of a verification circuit, a standard multilinearity test, and the LFKN sum-check protocol written out as a static oracle. The resulting proof has quasipolynomial length, which is harmless for the stated goal. We then obtain a PCP of proximity by one additional test: locally self-correct a uniformly random claimed input coordinate and compare it with the actual input oracle. The note is self-contained except for the standard multilinearity-testing theorem.

1 The statements

For strings $x, y \in \{0,1\}^n$, write

$$\operatorname{dist}(x,y) := \frac{1}{n}\big|\{i : x_i \neq y_i\}\big|, \qquad \operatorname{dist}(x,S) := \min_{y\in S} \operatorname{dist}(x,y),$$

with the convention $\operatorname{dist}(x,\varnothing) = 1$. The verifier is allowed polynomial running time; only its randomness and oracle-query complexity are being optimized. Proof symbols will initially lie in a finite field. Reading a field symbol costs $O(\log|\mathbb{F}|)$ Boolean queries, so the final systems are ordinary binary PCPs and PCPPs.

Theorem 1.1 (Weak algebraic PCP)**.** *For every language $L \in \mathsf{NP}$, there is a PCP verifier with perfect completeness and constant soundness that uses*

$$(\log n)^{O(1)}$$

random bits and Boolean proof queries on inputs of length n. The proof length is at most

$$\exp\big(O(\log^2 n)\big).$$

Theorem 1.2 (Weak algebraic PCPP)**.** *For every language $L \in \mathsf{NP}$, there is an oracle verifier V with perfect completeness such that, for every $x \in \{0,1\}^n$ and every alleged proof π,*

$$\operatorname{dist}(x, L\cap\{0,1\}^n) \geq \delta \quad\Longrightarrow\quad \Pr[V^{x,\pi} \text{ rejects}] \geq c\delta$$

for an absolute constant $c > 0$. It uses $(\log n)^{O(1)}$ random bits and Boolean queries, and the proof length is $\exp(O(\log^2 n))$.

Consequently, for every fixed constant $\delta_0 > 0$, this is a constant-soundness PCPP for inputs that are δ_0-far from L. Repeating $O(1/\delta)$ times gives constant soundness for a supplied proximity parameter δ.

Proof at a glance. Given a circuit and a purported accepting computation a:

$$\boxed{\begin{array}{c}\text{quadratic circuit constraints}\\ \Downarrow \text{ random fingerprint } \alpha\\ Q_\alpha(a)=0 \text{ (one quadratic identity)}\\ \Downarrow \text{ multilinear extension } P\\ \displaystyle\sum_{u,v\in\{0,1\}^m} F_{\alpha,P}(u,v)=0\\ \Downarrow \text{ tabulated LFKN sum-check}\\ \text{polylogarithmic verification.}\end{array}}$$

The PCPP adds only this: choose a random input coordinate i, recover the claimed value $P(u_i)$ from nearby random field points, and compare it with the actual oracle bit x_i.

2 The one black box: multilinearity testing

A polynomial $P : \mathbb{F}^m \to \mathbb{F}$ is *multilinear* if it has degree at most one in each variable. For functions $A, B : \mathbb{F}^m \to \mathbb{F}$, let

$$\operatorname{dist}(A, B) := \Pr_{z\in\mathbb{F}^m}[A(z) \neq B(z)].$$

We use the following standard consequence of the multilinearity test of Babai–Fortnow–Lund, in the formulation recorded by Babai–Fortnow–Levin–Szegedy [1, Theorem 4.4].

Theorem 2.1 (Multilinearity test)**.** *Let $\mathbb{F}$ be a finite field of order q. There is a randomized oracle test* MLT *for functions $A : \mathbb{F}^m \to \mathbb{F}$ with the following properties.*

1. *If A is multilinear, the test always accepts.*
2. *If A is not η-close to any multilinear polynomial, where*

$$\eta := \frac{4m^2}{q},$$

   then the test rejects with probability at least $9/10$*.*
3. *The test makes $m^{O(1)}$ field queries and uses $(m \log q)^{O(1)}$ random bits.*

The constant $9/10$ is obtained from the usual "high probability" formulation by constant repetition. We use no robustness statement: the dichotomy "far and detected, or close to one multilinear polynomial" is enough.

Remark 2.2. This is the sole non-elementary input. The rest of the note proves the arithmetization, fingerprinting, sum-check, static-oracle conversion, and PCPP self-correction directly.

3 Sum-check as a static proof oracle

We record the familiar LFKN sum-check protocol [2] in precisely the form needed later. The only mild observation is that an interactive prover strategy can be written down as a static table indexed by the public transcript.

Lemma 3.1 (Tabulated sum-check)**.** *Let $f : \mathbb{F}^k \to \mathbb{F}$ be a polynomial of degree at most d in each variable, and let $S \in \mathbb{F}$. There is a static proof oracle whose verifier tests*

$$S \stackrel{?}{=} \sum_{z \in \{0,1\}^k} f(z)$$

using k uniformly random field elements, $O(kd)$ queried field symbols, and one evaluation of f at a random point. It has perfect completeness and, when the displayed identity is false, acceptance probability at most

$$\frac{kd}{q}.$$

The proof table has at most $q^{O(k)}$ field symbols.

Proof. For $i = 1, \ldots, k$ and a prefix $r_{<i} = (r_1, \ldots, r_{i-1}) \in \mathbb{F}^{i-1}$, the proof table supplies the coefficients of a univariate polynomial

$$g_{i,r_{<i}}(T) \in \mathbb{F}[T], \qquad \deg g_{i,r_{<i}} \le d.$$

For an honest proof,

$$g_{i,r_{<i}}(T) = \sum_{z_{i+1},\ldots,z_k \in \{0,1\}} f(r_1, \ldots, r_{i-1}, T, z_{i+1}, \ldots, z_k).$$

The verifier reads g_1, checks

$$S = g_1(0) + g_1(1),$$

chooses $r_1 \in \mathbb{F}$ uniformly, and continues. In round $i > 1$, it reads $g_{i,r_{<i}}$, checks

$$g_{i-1,r_{<i-1}}(r_{i-1}) = g_{i,r_{<i}}(0) + g_{i,r_{<i}}(1),$$

and then chooses $r_i \in \mathbb{F}$ uniformly. At the end it checks

$$g_{k,r_{<k}}(r_k) = f(r_1, \ldots, r_k).$$

For soundness, let $G_{i,r_{<i}}$ denote the honest partial-sum polynomial. If the initial claim is false, then after the first consistency check either the verifier has rejected or $g_1 \neq G_1$. Whenever $g_{i,r_{<i}} \neq G_{i,r_{<i}}$, the two degree-$d$ polynomials agree at the random point r_i with probability at most d/q. Unless this coincidence occurs, the next consistency equation either rejects immediately or propagates a false polynomial to the next round. A union bound over k rounds gives error at most kd/q.

There are q^{i-1} possible prefixes in round i, and each polynomial uses $d + 1$ coefficients, so the whole table has $q^{O(k)}$ symbols. □

4 Quadratic arithmetization and fingerprinting

Fix $L \in \mathsf{NP}$. Choose a polynomial-time relation $R(x, w)$ with

$$x \in L \quad \Longleftrightarrow \quad \exists w \; R(x, w) = 1.$$

For a fixed input length, let C be a polynomial-size Boolean circuit for this relation, using only AND and NOT gates. Introduce one variable for every input, witness, and internal wire. Over a field of characteristic two, impose:

- $y + xz = 0$ for an AND gate $y = x \wedge z$;
- $y + x + 1 = 0$ for a NOT gate $y = \neg x$;
- $t^2 + t = 0$ for every wire variable t;
- $y_{\text{out}} + 1 = 0$ for the output wire.

For the ordinary PCP, the instance bits x_i are hardwired constants. For the PCPP, they are replaced by claimed input variables z_i; the actual oracle x will be compared with z only in the proximity test.

Thus we obtain quadratic polynomials

$$e_1(a), \ldots, e_M(a) \in \mathbb{F},$$

where $a = (a_1, \ldots, a_K)$ lists all claimed variables. Here $K, M = n^{O(1)}$. Put

$$m := \lceil \log_2 K \rceil.$$

The constraints vanish simultaneously exactly when a describes an accepting computation.

Choose a field

$$\mathbb{F} = \mathbb{F}_{2^t}, \qquad n^B \le q := |\mathbb{F}| < 2n^B,$$

where the constant B is taken sufficiently large. In particular, throughout we may assume

$$\frac{M}{q} \le \frac{1}{100}, \qquad \frac{m}{q} \le \frac{1}{100}, \qquad \frac{4m^2(m+1)}{q} \le \frac{1}{100n}. \tag{1}$$

For the finitely many smaller input lengths, the verifier may be hardwired.

For $\alpha \in \mathbb{F}$, define the fingerprint

$$Q_\alpha(a) := \sum_{j=1}^{M} \alpha^{j-1} e_j(a).$$

Lemma 4.1 (Fingerprinting)**.** *If all constraints vanish, then $Q_\alpha(a) = 0$ for every α. If at least one constraint is nonzero, then*

$$\Pr_{\alpha \in \mathbb{F}}[Q_\alpha(a) = 0] \le \frac{M-1}{q}.$$

Proof. In the second case, $Q_\alpha(a)$ is a nonzero univariate polynomial in α of degree at most $M-1$, and hence has at most $M-1$ roots. □

5 Turning the fingerprint into one Boolean-cube sum

Pad the vector a with unused coordinates so that it is indexed by $u \in \{0,1\}^m$. Let

$$P : \mathbb{F}^m \to \mathbb{F}$$

be its unique multilinear extension. Thus $P(u) = a_u$ on the Boolean cube.

For fixed α, the fingerprint is a quadratic polynomial in the coordinates of a. Write it as

$$Q_\alpha(a) = c_\alpha + \sum_{u \in \{0,1\}^m} \ell_\alpha(u) a_u + \sum_{u,v \in \{0,1\}^m} b_\alpha(u,v) a_u a_v. \tag{2}$$

Let $L_\alpha : \mathbb{F}^m \to \mathbb{F}$ and $B_\alpha : \mathbb{F}^{2m} \to \mathbb{F}$ be the multilinear extensions of ℓ_α and b_α. Finally define the Boolean-point indicator

$$\chi_0(X) := \prod_{j=1}^{m} (1 - X_j),$$

which equals one at 0^m and zero at every other Boolean point.

Define

$$\begin{aligned} F_{\alpha,P}(X,Y) := B_\alpha(X,Y)P(X)P(Y) \\ + L_\alpha(X)P(X)\chi_0(Y) + c_\alpha\chi_0(X)\chi_0(Y). \end{aligned} \tag{3}$$

Every variable has degree at most two in $F_{\alpha,P}$. Moreover, by construction,

$$\sum_{u,v\in\{0,1\}^m} F_{\alpha,P}(u,v) = Q_\alpha(a). \tag{4}$$

The verifier can evaluate $B_\alpha(X,Y)$ and $L_\alpha(X)$ in polynomial time from the circuit, α, and the requested field point. We do not try to make this computation polylogarithmic; verifier running time is allowed to be polynomial.

6 The weak PCP

The proof consists of two field-valued oracles.

1. A table $A : \mathbb{F}^m \to \mathbb{F}$, purportedly the multilinear extension P of an accepting computation.
2. For every $\alpha \in \mathbb{F}$, every sum-check round $i \in [2m]$, and every challenge prefix $r_{<i} \in \mathbb{F}^{i-1}$, the three coefficients of the purported degree-two polynomial $g_{i,\alpha,r_{<i}}$.

The verifier chooses one of the following two branches with equal probability.

Multilinearity branch. Run MLT on A.

Algebraic branch. Choose $\alpha \in \mathbb{F}$ uniformly and invoke the tabulated sum-check verifier for the claim

$$0 = \sum_{u,v\in\{0,1\}^m} F_{\alpha,A}(u,v).$$

At the final random point $(r,s) \in \mathbb{F}^m \times \mathbb{F}^m$, evaluate the right side of (3) using the two queried values $A(r)$ and $A(s)$.

Proof of Theorem 1.1. If $x \in L$, take an accepting computation a, its multilinear extension P, and all honest partial-sum polynomials. Both branches accept for every random string, giving perfect completeness.

Now suppose $x \notin L$ and fix arbitrary proof oracles. Put

$$\eta := \frac{4m^2}{q}.$$

If A is not η-close to any multilinear polynomial, the multilinearity branch rejects with probability at least $9/10$, so the overall rejection probability is at least $9/20$.

Otherwise, fix a multilinear P with $\operatorname{dist}(A, P) \le \eta$, and let $a = P|_{\{0,1\}^m}$. Since the instance is false, a violates at least one circuit constraint. By Theorem 4.1, with probability at least $1 - M/q$ over α,

$$\sum_{u,v \in \{0,1\}^m} F_{\alpha,P}(u, v) = Q_\alpha(a) \neq 0.$$

For each such α, Theorem 3.1 with $k = 2m$ and $d = 2$ shows that a sum-check ending in the genuine value $F_{\alpha,P}(r, s)$ accepts with probability at most $4m/q$. Our verifier instead uses $A(r), A(s)$, but r and s are individually uniform in $\mathbb{F}^m$. Hence

$$\Pr[A(r) \neq P(r) \text{ or } A(s) \neq P(s)] \le 2\eta.$$

Therefore the algebraic branch rejects with probability at least

$$1 - \frac{M}{q} - \frac{4m}{q} - 2\eta,$$

which is larger than $9/10$ for our choice of q. The overall rejection probability is at least $9/20$.

It remains to count resources. We have $m = O(\log n)$ and $\log q = O(\log n)$. The multilinearity branch uses polylogarithmic randomness and queries by Theorem 2.1. The algebraic branch uses $O(m \log q) = O(\log^2 n)$ random bits and $O(m)$ field queries. Replacing each field symbol by its $O(\log q)$-bit representation preserves polylogarithmic Boolean query complexity.

The table A has q^m field symbols. The sum-check strategy has

$$O\left(\sum_{i=1}^{2m} q^i\right) = q^{O(m)}$$

field symbols. Since $q = n^{O(1)}$ and $m = O(\log n)$, the binary proof length is $\exp(O(\log^2 n))$. □

7 The PCPP augmentation

We now treat the input $x \in \{0,1\}^n$ as an oracle. The algebraic proof claims an accepting computation for some string $z \in \{0,1\}^n$, with the coordinates of z occupying designated Boolean-cube points

$$u_1, \dots, u_n \in \{0,1\}^m.$$

The multilinearity and algebraic branches are unchanged. We add a third, equally likely branch that checks that the claimed z is close to the actual oracle x.

The apparent problem is that directly reading $A(u_i)$ is unsound: the designated Boolean points form a negligible fraction of $\mathbb{F}^m$, so an oracle may corrupt precisely those points while remaining globally close to a multilinear polynomial. We therefore self-correct $P(u_i)$ from random points, using the standard polynomial self-correction idea [3]. This PCP-to-PCPP augmentation was suggested already in BFLS and emphasized in later PCPP constructions [4].

7.1 A simple local corrector

Fix distinct nonzero elements $t_1, \dots, t_{m+1} \in \mathbb{F}$.

Lemma 7.1 (Local correction at one point)**.** *Suppose $A : \mathbb{F}^m \to \mathbb{F}$ is η-close to a multilinear polynomial P. Given any $u \in \mathbb{F}^m$, the following procedure outputs $P(u)$ except with probability at most*

$$(m+1)(\eta + q^{-m}).$$

1. *Choose $v \in \mathbb{F}^m$ uniformly; if $v = 0$, replace it by the first standard basis vector e_1.*
2. *Query $A(u + t_j v)$ for $j = 1, \ldots, m + 1$.*
3. *Interpolate the unique degree-at-most-m univariate polynomial through these values and output its value at zero.*

Proof. The restriction

$$h(T) := P(u + Tv)$$

has degree at most m, because P is multilinear and hence has total degree at most m. Since the resulting direction v is always nonzero, the points $u + t_1 v, \ldots, u + t_{m+1} v$ are distinct. If every queried value of A agrees with P, interpolation recovers h and outputs $h(0) = P(u)$.

Before the replacement of $v = 0$, each $u + t_j v$ is uniform on $\mathbb{F}^m$. Replacing the single outcome $v = 0$ changes its distribution by total variation at most q^{-m}. Hence each query lands in the disagreement set with probability at most $\eta + q^{-m}$. A union bound proves the claim. □

7.2 The proximity branch

Choose $i \in [n]$ uniformly, apply Theorem 7.1 to recover the claimed value $P(u_i)$ from A, and reject unless the recovered field element equals the oracle bit $x_i \in \{0, 1\} \subseteq \mathbb{F}$.

Proof of Theorem 1.2. Completeness is immediate. If $x \in L$, choose $z = x$ and an accepting witness. Then $A = P$ exactly, local correction always returns $P(u_i) = x_i$, and all three branches accept.

For soundness, fix x and arbitrary proof oracles, and assume

$$\operatorname{dist}(x, L \cap \{0, 1\}^n) \geq \delta.$$

As before, let $\eta = 4m^2/q$.

If A is not η-close to a multilinear polynomial, the multilinearity branch gives a constant rejection probability. Otherwise fix P with $\operatorname{dist}(A, P) \leq \eta$ and put $a = P|_{\{0,1\}^m}$.

If a violates a circuit constraint, the algebraic branch rejects with constant probability by exactly the PCP analysis above. We may therefore consider the remaining case in which a satisfies every constraint. The Booleanity constraints imply that the designated coordinates define a Boolean string

$$z_i := P(u_i), \qquad i \in [n],$$

and the remaining coordinates supply a witness and accepting computation. Thus $z \in L \cap \{0, 1\}^n$. Consequently,

$$\Pr_{i \in [n]} [x_i \neq z_i] \geq \delta.$$

By Theorem 7.1, the correction procedure returns z_i except with probability at most

$$(m + 1)(\eta + q^{-m}) \leq \frac{1}{25n}$$

for our choice of q. Every positive relative Hamming distance is at least $1/n$, so whenever $\delta > 0$ this error is at most $\delta/25$. The proximity branch therefore rejects with probability at least $24\delta/25$.

Since the verifier chooses among three branches with equal probability, in all cases its rejection probability is at least $c\delta$ for an absolute constant $c > 0$ (and in the first two cases it is actually constant). The proximity branch uses $m + 1 = O(\log n)$ field queries and $O(m \log q) = O(\log^2 n)$ random bits. The proof oracle is unchanged, so the proof length remains $\exp(O(\log^2 n))$. □

8 What was deliberately avoided

The construction proves only the weak target $\mathsf{PCP}(\text{polylog}, \text{polylog})$ and its PCPP analogue. This lets us omit all of the machinery normally introduced to obtain the sharp PCP theorem:

- no proof composition;
- no parallel repetition;
- no long code or dictatorship test;
- no near-linear or polynomial proof-length optimization;
- no robustness analysis;
- no Dinur-style gap amplification.

The price is quasipolynomial proof length. To reduce the proof length to polynomial while retaining polylogarithmic verification, one packs the assignment into a grid H^m with $|H| = \text{polylog}\, n$ and $m \asymp \log n / \log\log n$, as in BFLS. That optimization is standard but obscures the basic architecture.

Conceptually, the construction is exactly

LFKN sum-check + one low-degree consistency theorem + local self-correction for proximity.

References

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